

kirchhoff's laws - practice problems

name _____

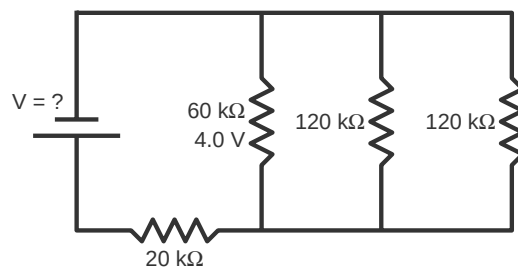
1. A loop has a 9.0 V battery and three voltage changes around the loop. Two resistor drops are -2.0 V and -3.5 V. What must the third voltage change be?
2. At a junction in a model train layout, 2.4 A flows in. Current leaves along three branches. Two branches carry 0.80 A and 0.60 A. How much current leaves in the third branch?
3. A circuit loop has a 9.0 V battery and two resistors labeled 100 Ω and 220 Ω . What is the current in the 220 Ω resistor?
4. Two resistors, 300 Ω and 600 Ω , are connected in parallel. What is their equivalent resistance?
5. You need 120 Ω for a circuit, but you only have a 300 Ω resistor. What resistor should you put in parallel with it?

6. A 12 V battery powers three series resistors: 100 Ω , 200 Ω , and 300 Ω . What is the circuit current, and what is the voltage across the 200 Ω resistor?

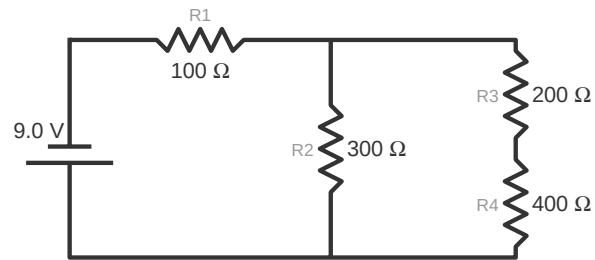
7. A 12 V power supply is connected across two parallel branches. One branch has a 2.0 k Ω resistor, and the other has a 3.0 k Ω resistor. What is the total current from the supply?

8. Two parallel branches are connected across a 9.0 V battery. One branch has a 1.0 k Ω resistor. The total current from the battery is 15 mA. What resistance is in the second branch?

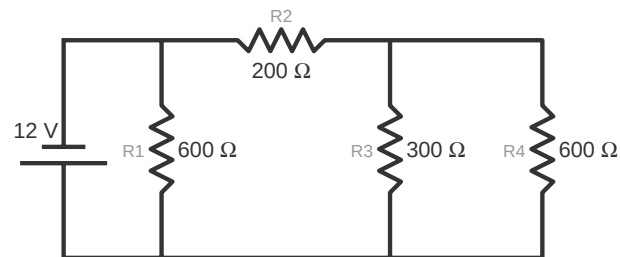
9. Find the voltage of the battery in the circuit below.



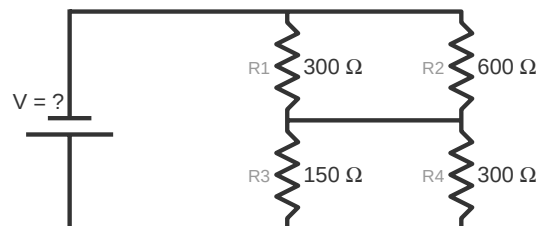
10. In the circuit below, R_3 and R_4 are in series with each other, and that pair is in parallel with R_2 . Find the current through R_4 and the voltage across R_4 .



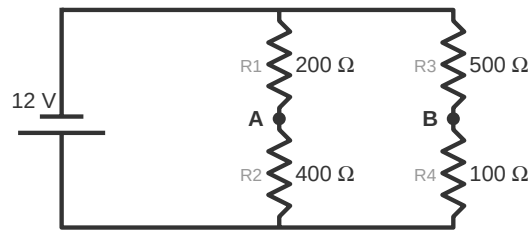
11. Find the current from the battery and the current through R_4 in the circuit below.



12. An ammeter shows 10 mA through R_2 in the circuit below. What is the battery voltage?



13. The bottom wire connects to the battery's positive terminal, the longer line, so let the top wire be at 0 V. What is the electric potential at point A and at point B? What is the potential difference between A and B?



14. A kitchen outlet circuit runs at 120 V. A toaster (12 Ω) and a microwave (14.4 Ω) are plugged in at the same time, so they are in parallel. The circuit breaker trips above 15 A. Does it trip?

15. A 9.0 V battery is connected to a 1.0 k Ω resistor in series with a parallel section. The parallel section has a 2.0 k Ω resistor and a 3.0 k Ω resistor. What is the total current?